Plannr

Timetable, attendance and scheduling for SICSR

GROUP MEMBERS

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COURSE

BCA Sem VII, Div A

Web UI and Content Management, Assessment 3

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plannr.paraswtf.com

The timetable is hard to use

- The college timetable is an MRBS install: PHP + MySQL, a LAMP application
- One desktop page for the whole college, plain HTTP, a 141 KB day view
- Coordinators book every lecture by hand
- Changes travel by Gmail, WhatsApp and Moodle

Research: 15 students, 10 faculty, 4 coordinators interviewed; 11 students surveyed.

73%

say it doesn't tell them when a class changes

82%

say finding their own classes takes too many steps

82%

can't see their week at a glance

87%

use WhatsApp to learn about changes

What we set out to do

01 A personal timetable: only your own classes, on any device

02 Clash-free scheduling with approvals in one queue

03 Attendance for every class, no setup per lecture

04 Changes follow into students' Google Calendar

05 Replace hand-booking with a timetable planner

06 Work with the college's existing MRBS data

Technology used

Next.js 15 · React 19 · TypeScript · tRPC · Prisma · PostgreSQL · Auth.js + Google · Tailwind CSS v4 · Vercel serverless · REST API + OpenAPI · Google Calendar API · Vitest

SYLLABUS CONCEPTS, AS APPLIED

Ajax
Every screen loads and saves with asynchronous JSON calls (tRPC, REST)

DOM
React updates the DOM; direct DOM APIs for swipe and measuring

Server-side database
Prisma + PostgreSQL in Node.js, the role PHP + MySQL play in LAMP

LAMP integration
Imports and reconciles the college's MRBS timetable, with a preview

The legacy version (web-ui-compliance branch) applies PHP, MySQL, Apache, jQuery, jQuery UI and Ajax directly.

Why Next.js, serverless and PostgreSQL

	Plannr's choice	LAMP, MySQL, WordPress
Kind of product	An application with scheduling logic	A CMS would need it all as custom plugins
Language	TypeScript from database to browser	PHP + jQuery splits it in two
Database	PostgreSQL: advisory locks, strict constraints	MySQL: not the locks and constraints Plannr relies on
Hosting	Vercel: HTTPS, preview per change, next to the DB	A server to patch and keep running
Today's proof	The college's LAMP timetable: one 141 KB page, desktop only, plain HTTP	

Built for this course: a legacy version (web-ui-compliance branch) re-creates the timetable viewer on LAMP with jQuery, jQuery UI and Ajax · github.com/paraswtf/plannr/tree/web-ui-compliance

Main features

Personal timetable
Day, week, month; swipe; opens on your next class

Google Calendar sync
Opt-in; moves and cancellations follow

Attendance
A register for every class; present, absent, late, excused

Requests and approvals
One queue showing clashes; locks stop double-booking

Clash detection
Room, teacher, group, capacity; next free slot

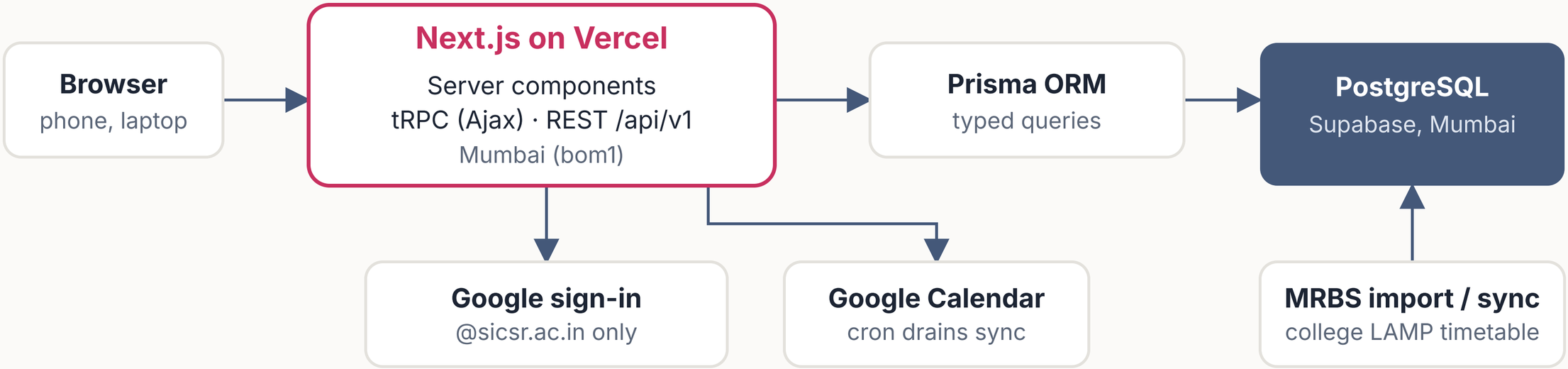
Room booking
Including recurring bookings

Timetable planner
Generates a clash-free week; review, approve, apply

Bulk CSV import
Preview every addition, change, removal

College sign-in and API
Google @sicsr.ac.in; REST API, OpenAPI, Siri

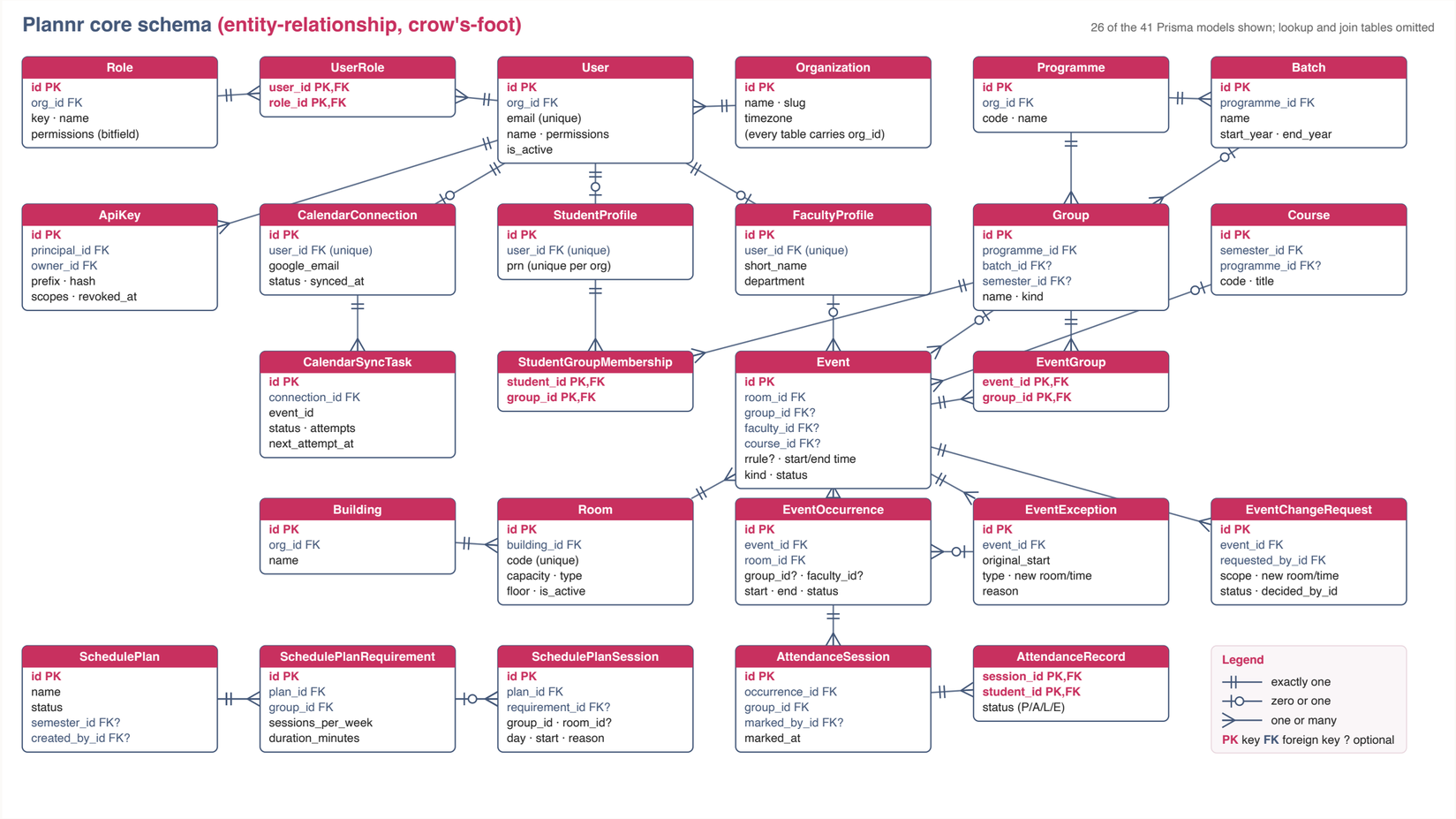
Architecture and workflow



REQUEST FLOW

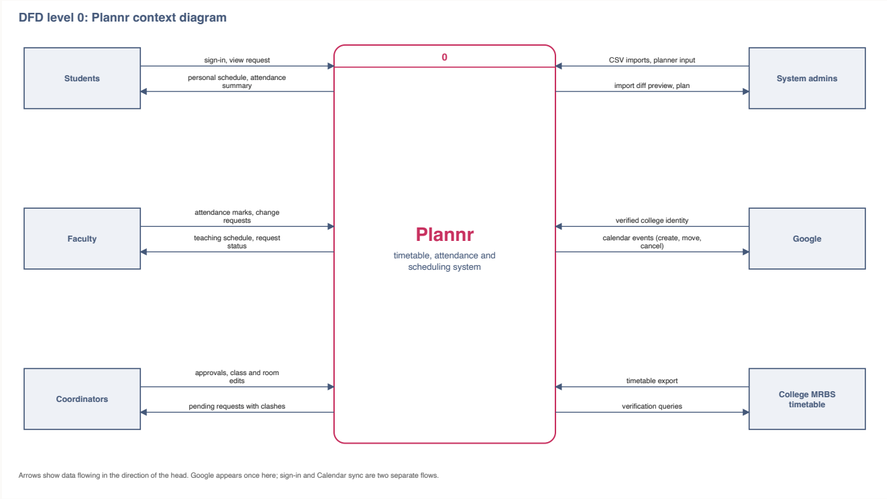


Data design: ER diagram

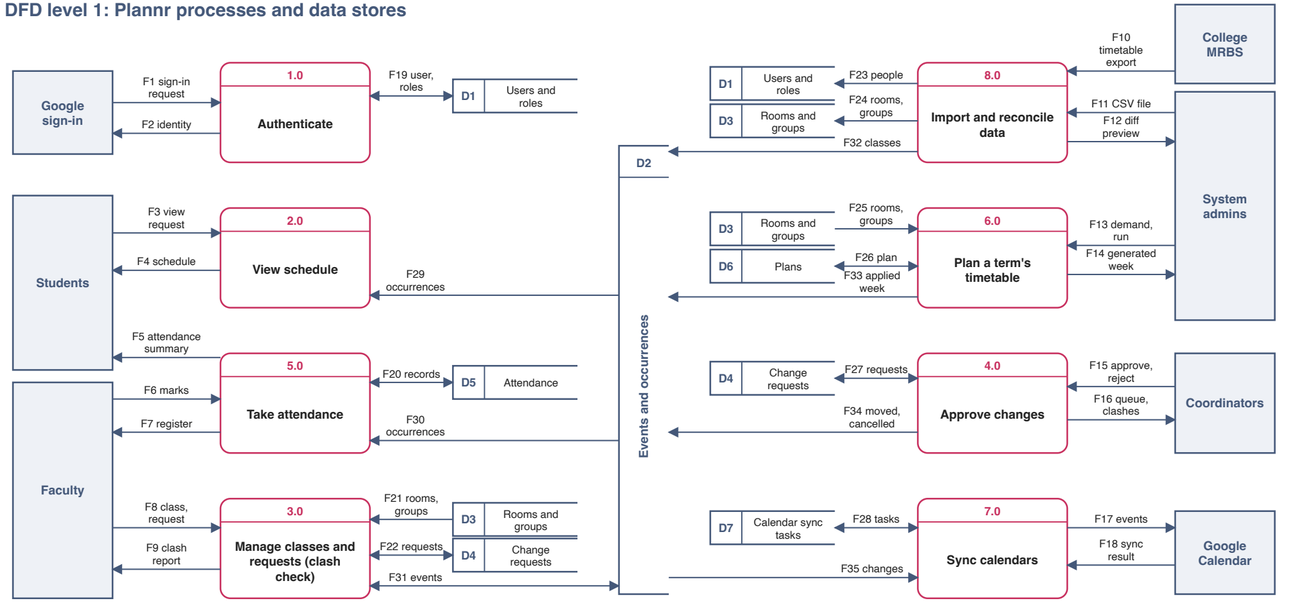


Data flow: DFD level 0 and level 1

LEVEL 0

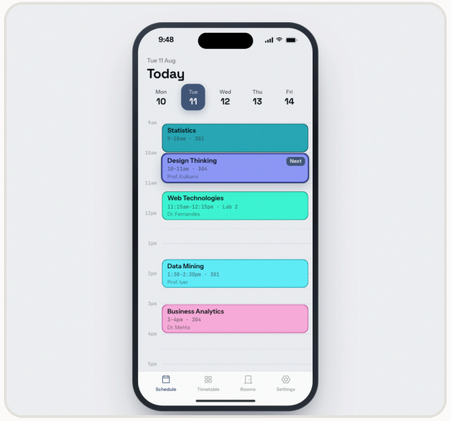


LEVEL 1

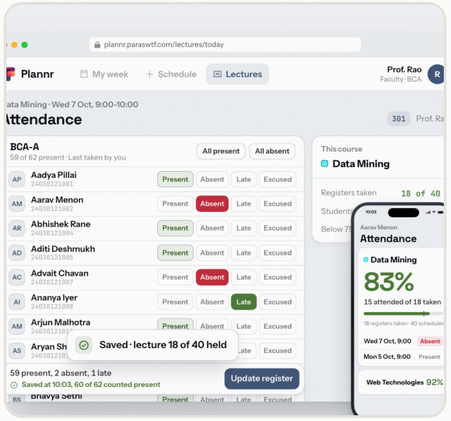


Flows F1 to F35 are labelled where they leave their source. D1, D3 and D4 are drawn more than once for readability. Double-headed arrows are read and write.
Students and faculty both read their personal schedule through process 2.0.

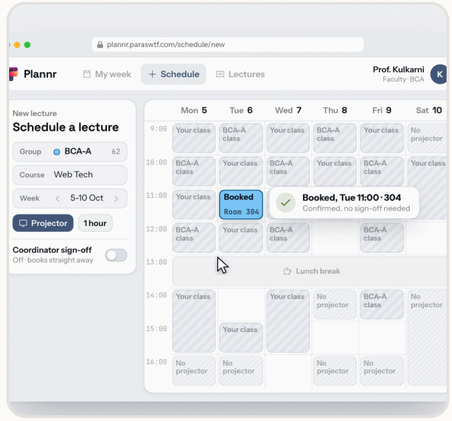
Plannr’s interface



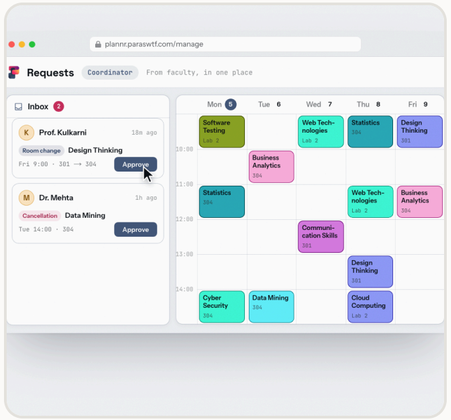
A student's day on a phone



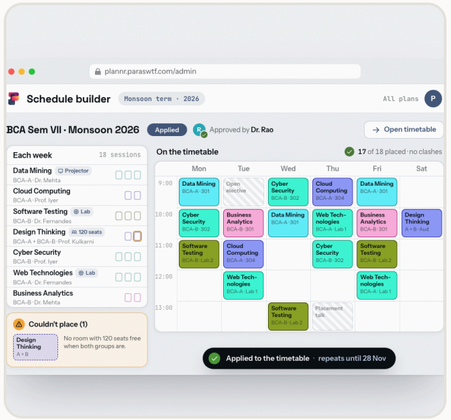
The attendance register



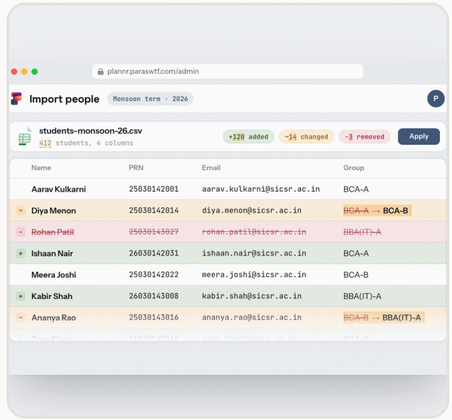
An extra lecture in a clash-free slot



The approvals queue



The timetable planner



CSV import with a diff preview

Demo and implementation

DEMO PATH · plannr.paraswtf.com

- Sign in with college Google
- Student: my day, week, attendance
- Faculty: request an extra lecture
- Coordinator: approve from the queue
- Admin: import CSV, run the planner

IMPLEMENTATION

- Recurrence materialised in one transaction
- Advisory locks stop double-booking
- Planner's solver: most constrained first, then repair
- 49 test files

Project files: **https://github.com/paraswtf/plannr**
Legacy version (LAMP, jQuery, Ajax): **https://github.com/paraswtf/plannr/tree/web-ui-compliance/web-ui-compliance**

Conclusion and future scope

CONCLUSION

Plannr turns the college's single desktop page into a personal timetable with attendance, approvals and a planner, built on the same data the college already holds.

FUTURE SCOPE

- Push, email and in-app change notifications
- Daily report of classes held
- Faculty availability; QR self check-in
- Tests and marks; assignments and deadlines
- Course material uploads; admin dashboard, audit log
- Offline mode